

Sameer Asif

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SUMMARY

Software and AI engineer and final-year BS Artificial Intelligence student at FAST-NUCES, currently building and maintaining production systems at Infigo Solutions. Comfortable across the full stack, from data pipelines and web scraping to backend APIs, databases, and machine learning models, with a focus on shipping work from prototype to deployment.

TECHNICAL SKILLS

- **Languages:** Python, SQL, C/C++, JavaScript, TypeScript, Bash
- **Backend & Data:** FastAPI, REST APIs, WebSockets, async Python, Playwright, SQL Server, PostgreSQL, SQLAlchemy, pandas, NumPy
- **Machine Learning:** PyTorch, Transformers, CNNs, GANs, computer vision, NLP, retrieval-augmented generation (RAG), ONNX
- **Frontend & Tools:** React, Vite, Git, Docker, Linux/WSL, Vercel, CUDA

EXPERIENCE

Junior AI Engineer | Infigo Solutions, Islamabad | July 2025 – Present

- Lead developer of the company’s customer-support assistant, a multi-channel application (website, WhatsApp, and Slack) built with FastAPI and PostgreSQL and deployed on Vercel, now running live on infigosolutions.com.
- Designed the system that answers customer questions from the company knowledge base and ERP records, including caching, usage limits, and safeguards to keep running costs low and responses accurate.
- Built an embeddable chat widget and public API for the website, and wrote the database migrations, seed data, and tests used across releases.

AI Engineering Intern | Infigo Solutions, Islamabad | June 2025 – July 2025

- Built a web-scraping pipeline (Clutch, Wellfound) that collected company and contact information at scale and loaded it into a normalized SQL Server database used by the sales team for lead generation.
- Added anti-bot handling and resumable progress tracking to keep long scraping runs reliable, plus regex-based extraction of emails, phone numbers, and LinkedIn profiles.
- Developed the first prototype of the support chatbot and set up its initial deployment and authentication, which led to a full-time offer.

PROJECTS

Adaptive Psychological Horror Game — Final Year Project | [GitHub](#)

- Trained facial-expression recognition models on several public datasets and combined video, audio, and heart-rate signals into a single stress classifier that drives a Unity game.
- Built generative models (a parameter generator, an audio GAN, and a texture network) that adapt the game’s sound and visuals to player stress in real time, exported to ONNX for use in Unity.

Voice-Cloning Conversational Agent | [GitHub](#)

- Built a real-time voice chat system that transcribes speech, generates replies with a language model, and responds in a cloned voice, running locally on a single 16 GB GPU.
- FastAPI backend with WebSocket streaming and a React frontend for voice upload and live playback.

Hybrid Wildlife Detection | [GitHub](#)

- Built a compact image detector that classifies animals and locates them in camera-trap photos across 21 species, combining techniques from three recent research papers and tuned to run on a 6 GB GPU.

Transformer and RAG from Scratch | [GitHub](#)

- Implemented a Transformer encoder and decoder from scratch in PyTorch for review sentiment classification (82% accuracy) and explanation generation, with a retrieval step over similar examples.

Other: [Urdu text generator](#) (statistical language models with a Streamlit app); [Sentinel trading bot](#) (automated Binance trading with sentiment signals); [SIP investment calculator](#) (React).

EDUCATION

BS Artificial Intelligence | FAST-NUCES, Islamabad | 2022 – 2026 (Expected) Relevant coursework: Deep Learning, NLP, Computer Vision, Databases, Software Engineering, Data Structures & Algorithms, Probability & Statistics

Cambridge A Levels | Beaconhouse School System | 2020 – 2022 | **Cambridge O Levels** | Beaconhouse | 2017 – 2020